

IB Mathematics: Analysis and Approaches

Standard Level – Paper 1

8 May 2023 – No Calculator

80 marks total

Section A

1. **[5 marks]** Point P has coordinates $(-3, 2)$, and point Q has coordinates $(15, -8)$. Point M is the midpoint of [PQ].
 - (a) Find the coordinates of M.
Line L is perpendicular to [PQ] and passes through M.
 - (b) Find the gradient of L .
 - (c) Hence, write down the equation of L .

2. **[7 marks]** The function f is defined by $f(x) = \frac{7x+7}{2x-4}$ for $x \in \mathbb{R}$, $x \neq 2$.
 - (a) Find the zero of $f(x)$.
 - (b) For the graph of $y = f(x)$, write down the equation of
 - (i) the vertical asymptote;
 - (ii) the horizontal asymptote.
 - (c) Find $f^{-1}(x)$, the inverse function of $f(x)$.

3. **[6 marks]** On a Monday at an amusement park, a sample of 40 visitors was randomly selected as they were leaving the park. They were asked how many times that day they had been on a ride called *The Dragon*. This information is summarized in the following frequency table.

Number of times on <i>The Dragon</i>	Frequency
0	6
1	16
2	13
3	2
4	3

It can be assumed that this sample is representative of all visitors to the park for the following day.

- (a) For the following day, Tuesday, estimate
- (i) the probability that a randomly selected visitor will ride *The Dragon*;
 - (ii) the expected number of times a visitor will ride *The Dragon*.

It is known that 1000 visitors will attend the amusement park on Tuesday. *The Dragon* can carry a maximum of 10 people each time it runs.

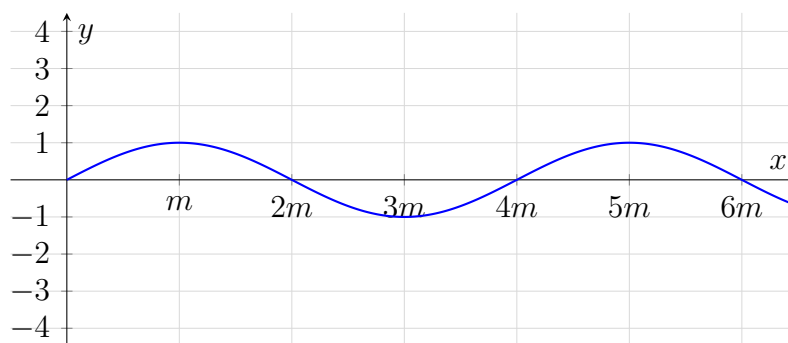
- (b) Estimate the minimum number of times *The Dragon* must run to satisfy demand.

4. [6 marks]

- (a) Show that the equation $\cos 2x = \sin x$ can be written in the form $2 \sin^2 x + \sin x - 1 = 0$.
- (b) Hence, solve $\cos 2x = \sin x$, where $-\pi \leq x \leq \pi$.

5. [6 marks] Find the range of possible values of k such that $e^{2x} + \ln k = 3e^x$ has at least one real solution.

6. [6 marks] The function f is defined by $f(x) = \sin qx$, where $q > 0$. The following diagram shows part of the graph of f for $0 \leq x \leq 4m$, where x is in radians. There are x -intercepts at $x = 0, 2m$ and $4m$.



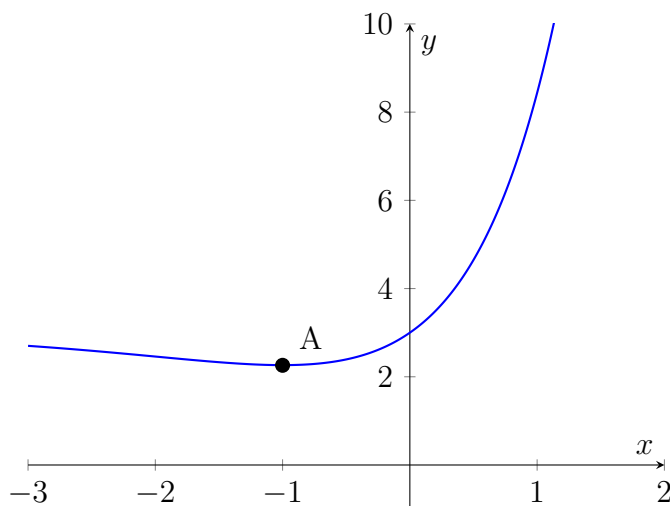
- (a) Find an expression for m in terms of q .

The function g is defined by $g(x) = 3 \sin \frac{2qx}{3}$, for $0 \leq x \leq 6m$.

- (b) On the axes above, sketch the graph of g .

Section B

7. [13 marks] The function h is defined by $h(x) = 2xe^x + 3$, for $x \in \mathbb{R}$. The following diagram shows part of the graph of h , which has a local minimum at point A.



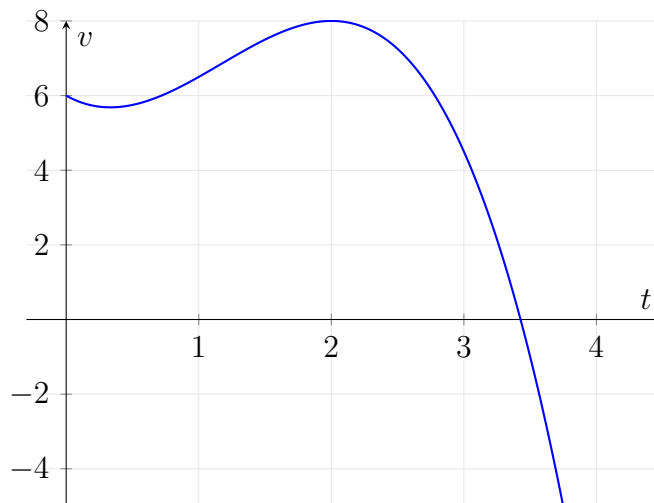
- (a) Find the value of the y -intercept.
 - (b) Find $h'(x)$.
 - (c) Hence, find the coordinates of A.
 - (d)
 - (i) Show that $h''(x) = (2x + 4)e^x$.
 - (ii) Find the values of x for which the graph of h is concave-up.
8. [14 marks] Consider the arithmetic sequence $u_1, u_2, u_3, \dots$.
The sum of the first n terms of this sequence is given by $S_n = n^2 + 4n$.
- (a)
 - (i) Find the sum of the first five terms.
 - (ii) Given that $S_6 = 60$, find u_6 .
 - (b) Find u_1 .
 - (c) Hence or otherwise, write an expression for u_n in terms of n .
- Consider a geometric sequence, v_n , where $v_2 = u_1$ and $v_4 = u_6$.
- (d) Find the possible values of the common ratio, r .
 - (e) Given that $v_{99} < 0$, find v_5 .

9. [17 marks] An object moves along a straight line. Its velocity, $v \text{ ms}^{-1}$, at time t seconds is given by

$$v(t) = -t^3 + \frac{7}{2}t^2 - 2t + 6, \text{ for } 0 \leq t \leq 4.$$

The object first comes to rest at $t = k$.

The graph of v is shown in the following diagram.



At $t = 0$, the object is at the origin.

- (a) Find the displacement of the object from the origin at $t = 1$.
- (b) Find an expression for the acceleration of the object.
- (c) Hence, find the greatest speed reached by the object before it comes to rest.
- (d) Find the greatest speed reached by the object for $0 \leq t \leq 4$.
- (e) Write down an expression that represents the distance travelled by the object while its speed is increasing. Do not evaluate the expression.